Lesson plan · for the teacher

How machines learn from examples

AI · Level 1 · Lesson 2

Time
40 min

Materials
Board and chalk (or large paper on the wall); Your plan of 18 shapes to draw; Worksheet: Teach the machine (one per pair, optional)

Learning goals

- Put the five steps of machine learning in order.
- Find a hidden pattern in labelled examples and use it to predict new ones.
- Explain how a spam filter learns from labelled messages.

Activity: Be the machine: Zog game

1. Warm-up (5 min): Draw a ripe and an unripe mango on the board (or hold up real ones). Ask pupils how they would teach a younger child to tell them apart.
2. Teach (8 min): Draw the five steps with simple icons: collect examples → label them → train: find patterns → test on new examples → use: make predictions. Use a spam filter as the example.
3. Zog game (14 min): Draw 6 zogs and 6 not zogs in two columns. Pupils study them silently for 2 minutes (training). Then draw the 6 test shapes one by one; pupils show 1 finger for zog, 2 for not zog. Keep score.
4. Pair talk (7 min): Pairs share the pattern they found. Take three ideas, then reveal the secret: zogs have three corners. Ask which shapes were hardest and what extra examples would help.
5. Wrap-up (6 min): Link the game to the five steps: the labelled drawings were the training data, and the new shapes were the test. Then give the exit ticket.

Secret pattern: zogs have three corners. Plan 6 zogs, 6 not zogs (squares, circles, stars) and 6 test shapes, including an upside-down triangle. Vary size, shading and direction in both groups, so only the corners matter.

Discussion

- To teach a young child which mangoes are ripe, would you give a rule, or show many mangoes and say ‘ripe’ or ‘not ripe’?
- Which examples helped you most?
- What new examples would help a machine that got this wrong?

Exit ticket

- Write the five steps in order, and one thing a spam filter learns from.

Try it in class tomorrow
Play “Be the machine”: On the board, draw six “zogs” (secret pattern: three corners) and six “not zogs”. Pupils study them — that’s training. Draw six new shapes; pupils predict “zog” or “not zog” — that’s testing. Reveal the pattern, then ask: how is this like a spam filter learning?

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Put the steps of machine learning in order.**

- ☐ Train: find patterns
- ☐ Use: make predictions
- ☐ Test on new examples
- ☐ Collect examples
- ☐ Label them

2 **You tap "Report spam" on a junk email. What does that give the spam filter?**

- ◯ A new rule that you wrote
- ◯ A finished model
- ◯ A prediction about the email
- ◯ A new example, labelled "spam"

3 **Every AI app keeps learning from you, all the time.**

- ◯ True
- ◯ False

4 **Match each tool to the examples it learned from.**

Spam filter	• • • •	Leaf photos labelled by experts
Cassava disease app	• • • •	Records from thousands of river gauges
Voice typing	• • • •	Emails labelled "spam" or "not spam"
Flood forecasts	• • • •	Recordings of native speakers

5 **Complete the sentence.**

We train a _____ on labelled examples, then _____ it on new ones — and more _____ examples usually help.

Word bank: rule · memorise · test · model · varied

Answers · for the teacher

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1 **Top to bottom: 3, 5, 4, 1, 2**

Examples and labels come first. Training builds the model, a test on new examples checks it, and then it is used to make predictions.

2 **A new example, labelled "spam"**

Your tap labels a real message "spam": one more labelled example for the filter to learn from.

3 **False**

Some tools are updated with new examples, but many models stay the same until the company retrains them.

4 **Spam filter — Emails labelled "spam" or "not spam"**
Cassava disease app — Leaf photos labelled by experts
Voice typing — Recordings of native speakers
Flood forecasts — Records from thousands of river gauges

Each tool learned from its own kind of examples.

5 **We train a model on labelled examples, then test it on new ones — and more varied examples usually help.**

Train a model, test it on new examples — and when it makes mistakes, more varied examples usually help.

For you to think about

What could your pupils teach a machine to sort in your subject? Which examples and labels would it need?